

THE HIPEC PROCEDURE

What Patients Should Know Quick Tips – Care Guide



Key Takeaway

Hyperthermic Intraperitoneal Chemotherapy (HIPEC) is being explored as a treatment option for non-uterine leiomyosarcoma (LMS), particularly in cases where the cancer has spread within the abdominal cavity. This approach combines cytoreductive surgery with heated chemotherapy to target residual cancer cells, potentially improving outcomes for patients with advanced disease.

HIPEC is not currently a standard treatment for uterine leiomyosarcoma (uLMS), but new research suggests it may deserve consideration for a very select group of patients.

Historically, HIPEC has not been routinely recommended for uLMS. However, a small prospective Phase II study from Stanford reported encouraging results in 2026 for patients with **recurrent uLMS that could be completely removed surgically**.

For now, HIPEC should still be considered **investigational for uLMS—not an established standard of care**.

What is HIPEC?

HIPEC stands for Hyperthermic Intraperitoneal Chemotherapy.

HIPEC is performed during major abdominal surgery:

1. Surgeons first remove as much visible cancer as possible through **cytoreductive surgery**.
2. Heated chemotherapy is then circulated throughout the abdominal cavity.
3. The goal is to expose microscopic cancer cells remaining on the abdominal or **peritoneal surfaces** to a high concentration of chemotherapy.

Important: HIPEC treats the abdominal/peritoneal cavity. It does **not** treat cancer cells elsewhere in the body.

Overview of HIPEC for Non-Uterine LMS

HIPEC is gaining attention for its potential benefits in treating non-uterine LMS.

- **Treatment Approach** — HIPEC involves administering heated chemotherapy directly into the abdominal cavity after cytoreductive surgery, aiming to eliminate remaining cancer cells.
- **Efficacy Studies** — Recent studies indicate that HIPEC may improve survival rates and reduce recurrence in patients with advanced non-uterine LMS.
- **Safety Profile** — While generally well-tolerated, HIPEC can have side effects, including abdominal pain and infection, which need to be monitored.
- **Clinical Trials** — Ongoing clinical trials are assessing the effectiveness of HIPEC combined with various chemotherapy agents for non-uterine LMS.
- **Multidisciplinary Approach** — Successful treatment often requires a team of specialists, including surgical oncologists and medical oncologists, to optimize patient outcomes

Why Isn't HIPEC Standard for uLMS?

Uterine leiomyosarcoma can spread differently from cancers for which HIPEC is more established.

uLMS can spread **through the bloodstream** to distant areas such as the:

- Lungs
- Liver
- Bone
- Other organs

Because HIPEC works only within the abdominal cavity, it cannot treat microscopic or established disease outside that area.

Another unanswered question is whether HIPEC provides significant benefit **beyond complete surgical removal of the cancer itself**. Patients selected for HIPEC often have limited disease that surgeons believe can be completely removed, and complete cytoreduction may itself be an important factor in improved outcomes.

What Changed in 2026?

Researchers at Stanford reported results from a prospective **Phase II study specifically involving recurrent uLMS**.

Study highlights:

- **17 women** with recurrent uLMS participated.
- All underwent complete cytoreductive surgery.
- All visible disease was removed (**CCR-0**).
- HIPEC was performed using **gemcitabine**.
- Patients later received systemic **dacarbazine**.
- Median progression-free survival was reported as **11 months**, compared with the study's historical benchmark of 3 months.
- Median overall survival had not been reached after more than three years of follow-up.

These findings are encouraging—but they require caution.

The study was **small, conducted at a single center, and did not include a randomized control group**. Because every patient also underwent complete surgery and received additional systemic chemotherapy, researchers cannot yet determine how much benefit came specifically from HIPEC.

Bottom line: The study provides an important reason to continue researching HIPEC in uLMS, but it does not establish HIPEC as standard treatment.

5 Questions to Ask Your Sarcoma Team

1. Is my recurrent uLMS limited to the abdomen/peritoneum, or is there disease elsewhere?
2. Could all visible abdominal disease realistically be removed surgically?
3. Would cytoreductive surgery itself be appropriate for my situation?
4. Is there a reason to consider HIPEC at a center with specific uLMS expertise or through a clinical study?
5. What evidence suggests HIPEC would provide additional benefit beyond surgery alone in my particular case?

REMEMBER:

HIPEC is not simply another type of chemotherapy. It is performed as part of a major abdominal operation and can add treatment burden, recovery time, and potential complications.

If a sarcoma surgeon says HIPEC is not an option for uLMS, that does **not** mean they are overlooking an established treatment. Their recommendation is consistent with the fact that HIPEC is **not currently standard care for uLMS**.

The newer research does, however, give patients and physicians a reason to **continue watching this evolving area of research** and, in carefully selected cases, consider seeking an expert opinion.

Resources

Stanford Phase II Study: [Cytoreductive surgery and HIPEC for recurrent uterine leiomyosarcoma Clinical Trial: NCT04727242](#)

Efficacy and Safety of Cytoreductive Surgery with Hyperthermic Intraperitoneal Chemotherapy for Recurrent Uterine Leiomyosarcoma: [Results of a Phase 2 Study](#)



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